SSM Optimization

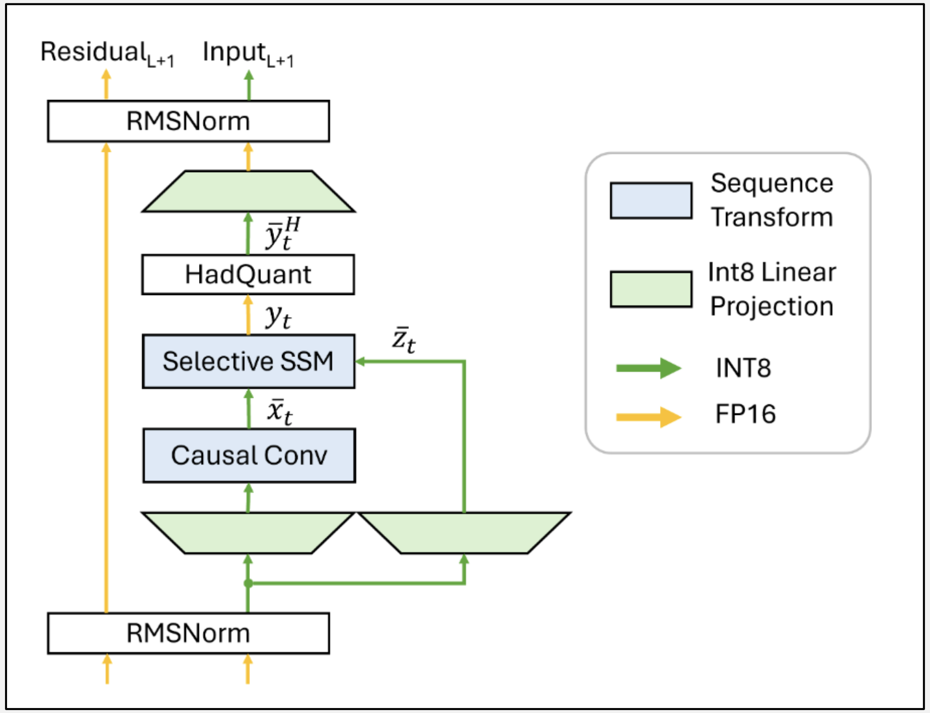
Weekly Meeting

01/06/26

Beatrice Alessandra Motetti

Quamba

- Static scaling factors computed from a calibration dataset
- **Clipping** on SSM block's input (p = 99.999)
- **Hadamard matrices** on the SSM block's output



MambaQuant

- W8A8 quantization, two modes according if the rotation matrix is integrated into the weights matrices:
  - Offline (no integr.): multiplies the Hadamard matrix with the **Karhunen-Loève Transformation** matrix, that equalizes variances in the channels (for gate, state and output proj.)

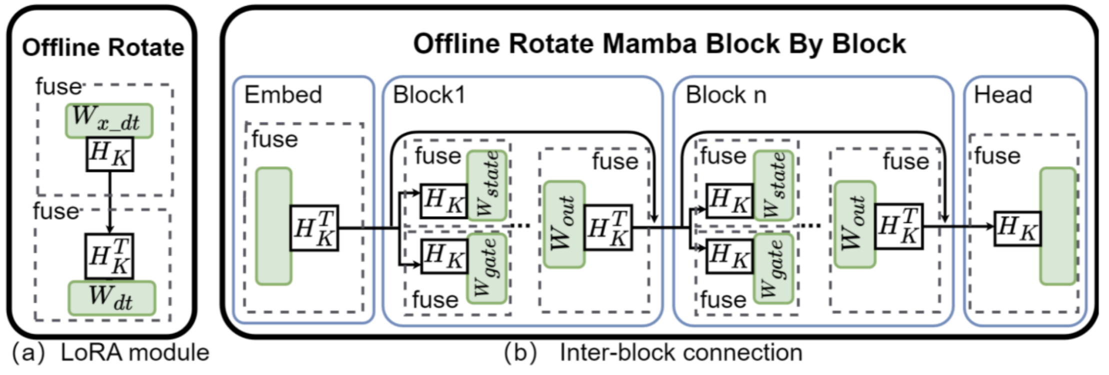


Figure 4: Offline transformation designs utilizing the KLT-Enhanced rotation.

  - Online (integr.): performs **smoothing** before the Hadamard transformation, with smoothing parameters integrated into weights

Q-Mamba

- Quantize **linear layers** of Mamba2 models and **state caches** (idea is to load and dequantize them before computation)
- **Separate scales** for the state and the channel dimensions
- They **fine-tune the linear layers** (except input and output projection layers) on 128 samples of length of 2048. They also apply SmoothQuant with per-token quantization

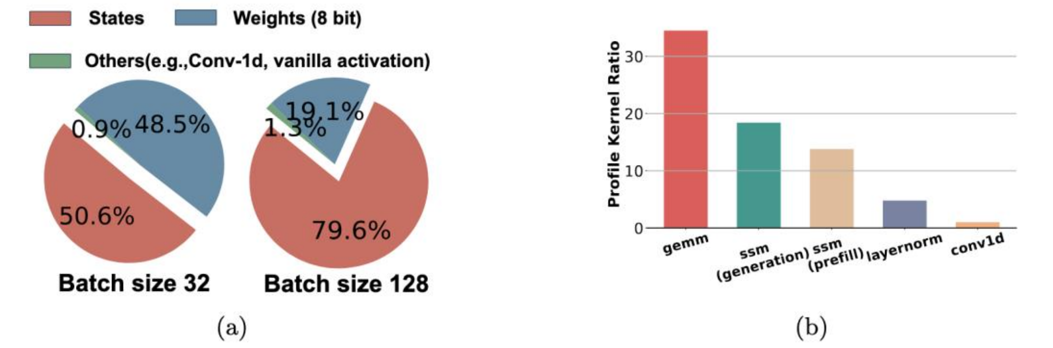


Figure 2: **Left**: Memory consumption of weights and state caches in Mamba2-2.7B with different batch sizes. **Right**: The Runtime of the Mamba2-2.7B model using NVIDIA profiling tools, with both prompt and generation lengths set to 100 and a batch size of 32.

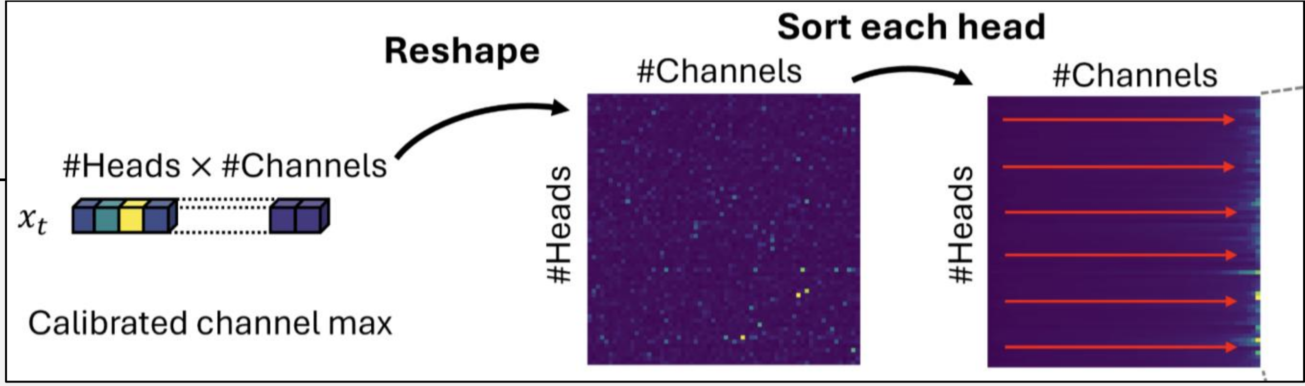
Quamba2

- PTQ method with static, pre-calibrated scaling factors

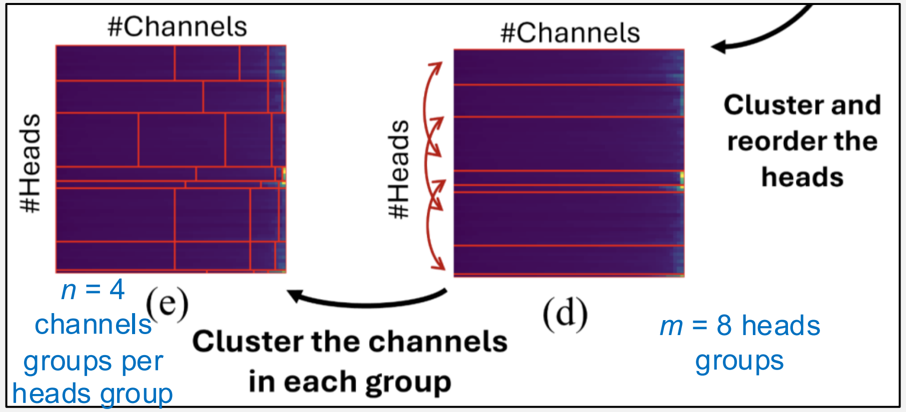
Methods	Models		Bitwidth		
	Mamba1	Mamba2	W8A8	W4A8	W4A16
MambaQuant (Xu et al.)	✓	-	✓	✓	-
Quamba (Chiang et al.)	✓	-	✓	-	-
Quamba2 (Ours)	✓	✓	✓	✓	✓

Methods	Embed.	SSM blocks	lm head	H2T
MambaQuant (Xu et al.)	16-bit	4/8-bit	16-bit	✗
Quamba (Chiang et al.)	16-bit	8-bit	16-bit	✗
Quamba2 (Ours)	4/8-bit	4/8-bit	4/8-bit	✓

Quamba2



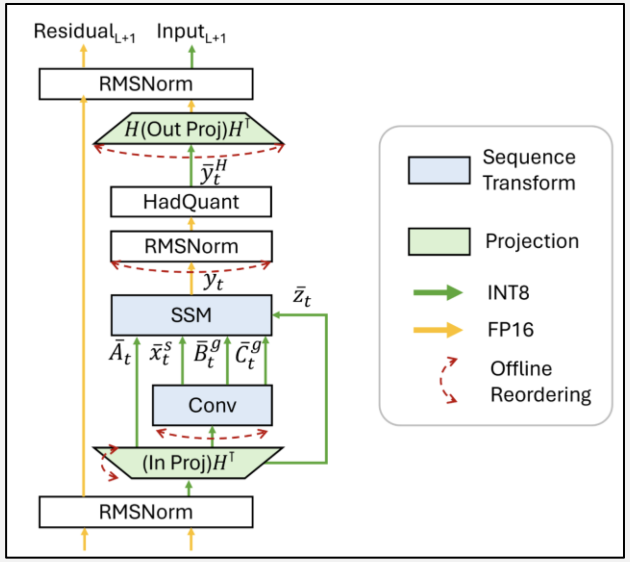
- **Sort-and-cluster**: start from the observation that the *magnitude of activations* in each channel is consistent across time steps and input samples
  - **Sort** (offline) the channels of *x* in descending order based on their calibrated maximum values
  - Apply **clustering** twice in a row, obtaining *m* x *n* scaling factors



Quamba2

- **Per-state group quantization**: B and C are the same for all channels of input x_t, and state channels with large activations are the same across time steps and input samples
  - Define groups in B and C and use a different scaling factor per group (e.g., 8 groups with 128 channels each for Mamba2-8B)

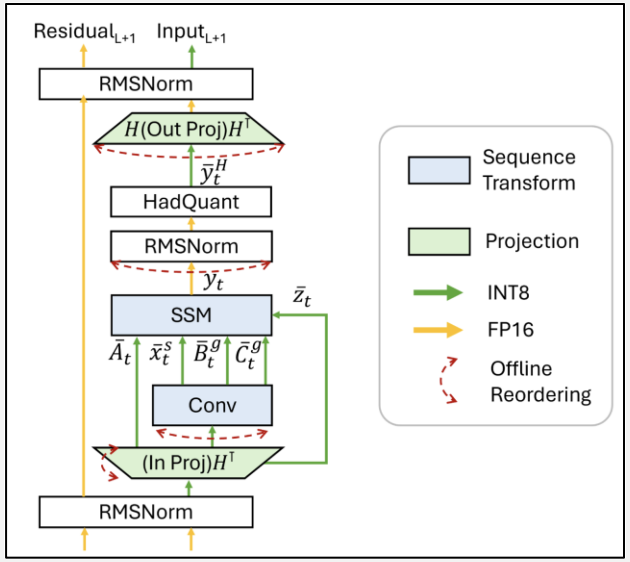
- **Rotations with Hadamard matrices**: fused within input and output linear layers



Quamba2

- **Per-state group quantization**: B and C are the same for all channels of input x_t , and state channels with large activations are the same across time steps and input samples
  - Define groups in B and C and use a different scaling factor per group (e.g., 8 groups with 128 channels each for Mamba2-8B)

- **Rotations with Hadamard matrices**: fused within input and output linear layers



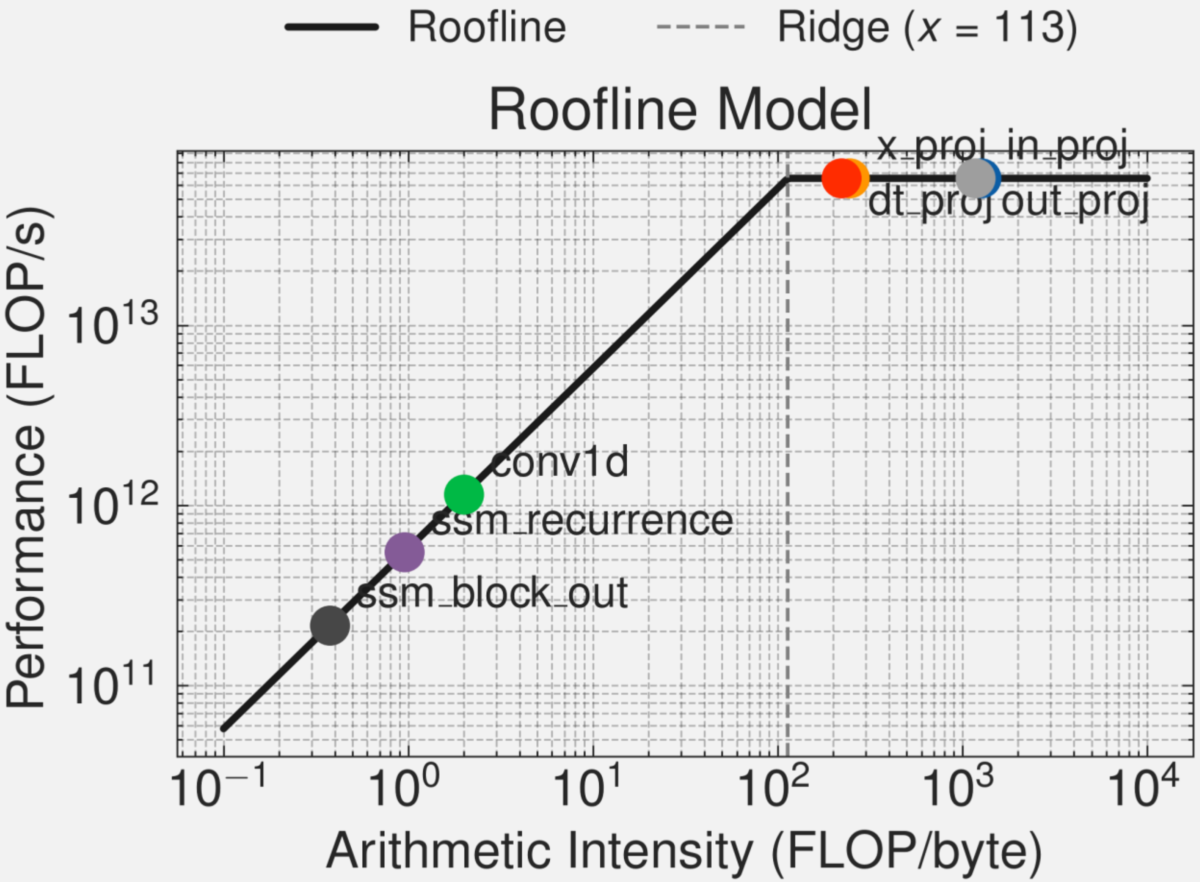
Slow forward

- Recurrent mode:

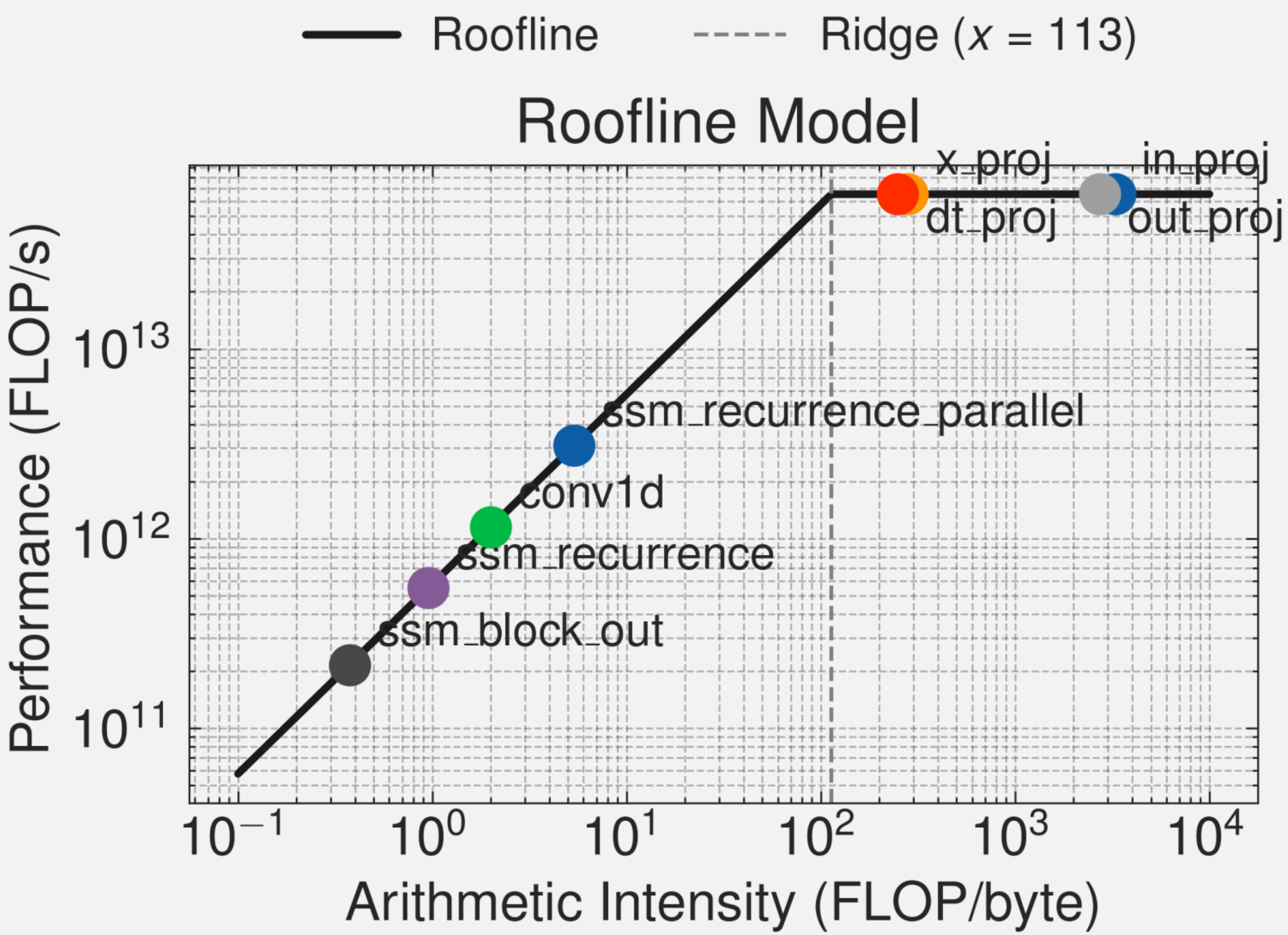
Block	AI
in_projection	$\frac{Bd_{int}S(2d_{inp}-1)}{b[BS(d_{int}+d_{inp})+2d_{inp}d_{int}]}$
conv1d	$\frac{2BSKd_{int}}{b[2BSd_{int}+d_{int}K]}$
x_proj	$\frac{Bd_{xp}S(2d_{int}-1)}{b[BS(d_{int}+d_{xp})+d_{xp}d_{int}]}$
dt_proj	$\frac{Bd_{int}S(2d_{dt}-1)}{b[BS(d_{dt}+d_{int})+d_{dt}d_{int}]}$
ssm_core_recurrence	$\frac{2BSd_{int}d_{ss}+BS(2d_{ss}-1)d_{int}}{b[2BSd_{int}d_{ss}+2Bd_{int}d_{ss}+BSd_{ss}+BSd_{int}]}$
ssm_block_out	$\frac{3BSd_{int}}{b[4BSd_{int}+d_{int}]}$
out_proj	$\frac{Bd_{inp}S(2d_{int}-1)}{b[BS(d_{inp}+d_{int})+d_{inp}d_{int}]}$

Slow forward

- Recurrent mode:



Complete


Roofline
Ridge (x = 113)
Roofline Model
x_proj
in_proj
dt_proj
out_proj
ssm_recurrence_parallel
conv1d
ssm_recurrence
ssm_block_out
Performance (FLOP/s)
10^{13}
10^{12}
10^{11}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Arithmetic Intensity (FLOP/byte)

Next steps

- Proceed to quantize activations (state in particular could be interesting), to try to overcome the memory-bound regime
  - Start with RTN.
  - Analyze sensitivity.